iPad + USB DSLR Photobooth Guide

Connect professional Canon, Nikon, and Sony DSLRs directly to your iPad via USB-C using Apple's native **ImageCaptureCore** framework for real studio optics, strobe flash sync, and 100% offline kiosk events.

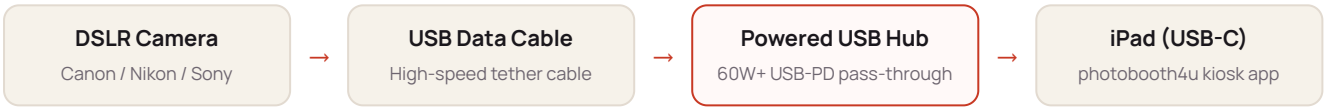
The Secret: Apple Native ImageCaptureCore (No WebUSB)

IPADOS 15.0+

Standard iPad browsers like Safari block direct USB access (WebUSB). **photobooth4u for iPad** bypasses this completely by running as a native iPadOS wrapper. It communicates directly with connected cameras through Apple's low-level **ImageCaptureCore PTP framework** on `127.0.0.1:17321`, delivering real-time live view, instant shutter triggering, and full-resolution photo downloads straight into memory.

1. The Single-Port iPad Hardware Wiring Chain

Because iPads have only one USB-C (or Lightning) port, you must route power and camera data through a **Powered USB-C Hub with Power Delivery (PD)** so the iPad never dies during a live event.



Essential Hardware Checklist

- **iPad with USB-C:** iPad Pro (any), iPad Air 4/5/M2, iPad 10th Gen, or iPad mini 6. (Lightning iPads require Apple Lightning to USB 3 Camera Adapter).
- **Powered USB-C Hub:** Must feature at least one USB-A/C data port AND a **USB-C Power Delivery (PD) In** port (Anker, Ugreen, Satechi).
- **High-Wattage Wall Charger:** 30W–65W USB-C wall adapter plugged into the hub's PD port to power both the iPad and attached peripherals.
- **Camera USB Data Cable:** Genuine camera manufacturer data cable (avoid charge-only phone cords).

Pro Event Tip: AC Dummy Battery

Do NOT rely on regular camera batteries at long weddings or corporate gigs. A DSLR in constant Live View drains a battery in 60–90 minutes.

> **Use a DC Coupler (Dummy Battery)** Plug a dummy battery directly into a wall outlet or high-capacity power bank. Your camera stays powered on for 12+ hours with zero battery swaps mid-event.

Camera Menu Configuration by Brand

Before plugging your camera into the iPad hub, set the camera dial and menu settings to allow external PTP host tethering control.

CANON EOS

Canon EOS DSLRs & Mirrorless

EOS R, R5, R6, R8, R10, R50, RP, 5D IV, 6D II, 90D, 80D, 200D/250D, M50 II

- **Dial:** Set top dial to **Manual (M)** mode.
- **Exposure:** Shutter 1/125s, f/5.6 – f/8, ISO 200–400 (with strobe).
- **USB Mode:** Set to **PC Remote** or **Computer**.
- **Wi-Fi/Bluetooth:** Turn **OFF** in the camera menu (avoids port lockout).
- **Auto Power Off:** Set to **Disable** so camera never sleeps.
- **SD Card:** Keep a formatted SD card inside camera.

NIKON

Nikon Z & D-Series

Z5, Z6, Z7, Z50, Zfc, D850, D780, D750, D7500, D5600, D3500, D3400

- **Dial:** Turn dial to **Manual (M)** mode.
- **Focus Mode:** Set to **AF-S (Single AF)** with wide-area or face detection.
- **USB Connection:** Set to **MTP / PTP** in Setup Menu.
- **Standby Timer:** Set c2 (Standby Timer) to **No Limit** or 30 mins.
- **Flash Sync:** Set built-in flash or hot-shoe speedlite to Manual.
- **Live View:** Enabled automatically upon iPad connection.

SONY ALPHA

Sony Alpha Mirrorless

A7 III, A7 IV, A7C, A7R, A6000, A6100, A6400, A6600, A6700, ZV-E10

- **Dial:** Set mode dial to **Manual (M)**.
- **USB Connection:** MENU → Setup → **PC Remote** (Crucial!).
- **USB LUN Setting:** Set to **Multi**.
- **PC Remote Settings:** Save Destination → **PC + Camera** (keeps SD backup).
- **Smartphone Control:** Set to **OFF** (required to allow USB tethering).
- **Power Save Start Time:** Set to **30 Min** or Off.

Studio Lighting & Strobe Flash Synchronization

Because the iPad triggers the physical shutter of your DSLR via PTP, you get **true hot-shoe flash synchronization**. Mount a Godox, Neewer, or Flashpoint wireless radio trigger (e.g., Godox Xpro or X2T) onto your camera's hot-shoe. When the countdown finishes on the iPad screen, the camera shutter fires and triggers your studio softboxes or beauty dishes with zero shutter lag.

Recommended Camera Exposure Settings for Photo Booths

Lighting Setup	Shutter Speed	Aperture	ISO	Focus Mode
Studio Strobes / Softbox	1/125s – 1/160s	f/6.3 – f/8.0	ISO 100 – 200	Single AF (AF-S) on backdrop center
LED Constant Ring Light	1/80s – 1/100s	f/3.5 – f/4.5	ISO 400 – 800	Face Detection / Continuous AF
Ambient / Indoor Venue	1/60s – 1/80s	f/2.8 – f/4.0	ISO 800 – 1600	Center Point AF with AF Assist Beam

iPadOS Kiosk Setup & USB Permissions

Configure your iPad settings to keep the screen awake, enable permanent USB accessory data access, and lock down the booth so guests cannot exit into your apps.

1. Prevent USB Disconnect (Crucial)

By default, iPadOS locks down its USB-C port after 60 minutes of inactivity to protect your device. For live events, you must disable this lock:

1. Open iPad **Settings**.
2. Tap **Face ID & Passcode** (or Touch ID & Passcode).
3. Scroll down to the *Allow Access When Locked* section.
4. Toggle **USB Accessories** to **ON (Green)**.

> Your DSLR will now stay permanently tethered without sudden mid-event disconnects.

2. Apple Guided Access (Kiosk Lock)

Lock the iPad into **photobooth4u** so guests cannot swipe home, view notifications, or change iPad settings:

1. Open **Settings → Accessibility → Guided Access**.
2. Turn **Guided Access** ON and set a 4-digit Passcode.
3. Open the **photobooth4u** app.
4. **Triple-click the Top Power Button**. Tap **Start** in the upper right.

> To exit after the event: Triple-click the Top button, enter your passcode, and tap **End**.

3. Launching & Testing in photobooth4u

Step A: Unlock Licence

Enter your photobooth4u code on the activation screen. The iPad needs Wi-Fi once to verify; thereafter it works 100% offline.

Step B: Select Camera

In the booth top bar, tap the Camera icon. Your attached Canon, Nikon, or Sony DSLR will appear in the list. Tap it to start Live View.

Step C: Test Capture

Tap the screen to trigger a countdown. The camera fires, the photo downloads to the iPad, and the photo strip preview renders instantly!

4. Local File & Photos Storage Architecture

Every photo taken on the iPad is automatically saved in two locations on device (zero cloud required):

1. In-App Event Storage

`Documents/Pictures/photobooth4u/<Event>/`

Full-resolution originals and finished formatted photo strips (2x6 or 4R).

2. iPad Photo Library (Camera Roll)

Apple Photos App → Recents

Easily AirDrop all event photos to your client's iPhone or Mac at the end of the night!

Local Wi-Fi QR Downloads & Printing

Guests scan an on-screen QR code to download their high-res photos and GIFs directly from the iPad over local Wi-Fi without needing cell service or cloud uploads.

How Local QR Downloads Work

The iPad app runs an embedded read-only HTTP server on **port 47777** (`LocalShareServer.swift`):

- **Step 1 (Join Wi-Fi):** Guests connect their smartphones to the same local Wi-Fi or pocket router as the iPad.
- **Step 2 (Scan Result QR):** At the end of each session, the screen displays a QR code linking to:

```
http://<ipad-ip>:47777/e/<event>/
```

- **Step 3 (Instant Save):** The guest's phone opens the branded mobile gallery with a **Save Photo** button.

> **Zero data usage:** All photos transfer locally over the venue router at up to 100 Mbps!

Wireless Printing from iPad

Print physical 4R photo cards or 2x6 strips instantly:

- **AirPrint Dye-Sub Printers:** Connect directly to Canon SELPHY (CP1300, CP1500), Epson PictureMate, or DNP/HiTi via wireless print servers (WPS Pro).
- **Receipt Photobooth:** Connect to thermal receipt printers over Bluetooth or network IPP.
- **Auto-Cut:** Set your layout to 2x6 dual strips; AirPrint sends the cutting command directly to the printer spooler.

> Turn on **Silent Auto-Print** in booth settings to skip all confirmation dialogs.

Pre-Event Operator Checklist (Print & Laminate)

Hardware Rig:

- [ ] iPad charger plugged into USB-C Hub PD port.
- [ ] Camera dummy battery plugged into wall power.
- [ ] High-speed USB data cable firmly secured.
- [ ] Wireless flash trigger powered on & test fired.

iPad & Camera Settings:

- [ ] Camera dial set to Manual (M) with focus locked.
- [ ] iPad Settings → USB Accessories turned **ON**.
- [ ] iPad volume set to 100% for voice prompts.
- [ ] Guided Access locked (Triple-click top button).

Troubleshooting: What If the Camera Won't Connect?

1. **Unplug & replug:** Unplug the USB-C cable from the iPad and plug it back in. iOS will re-initialize the `ICDeviceBrowser` session.
2. **Check USB mode:** On Sony, confirm USB is set to *PC Remote*. On Canon, turn off Wi-Fi/Bluetooth in camera menu.
3. **Check power:** Ensure the iPad hub's power LED is glowing. A hub without enough current cannot handshake with a DSLR sensor.